

CSCI 232 Assignment 03 – C Pointers & Node Fundamentals

Points: 10

Instructions

1. **Fork** the repository on GitHub: <https://github.com/jpach-cs/232-ass03-template>
2. **Clone** your forked repository to your local machine.
3. Open `src/code.c` and `src/tests.c`.
4. Uncomment the author block at the top of `src/code.c` and fill in your name:

```
// char *AUTHOR_NAME      = "Your Name";  
// char *AUTHOR_AUTHORSHP = "I acknowledge that I have worked on this  
// assignment independently...";
```

5. Work through each function in **two parts**:

Part 1 — `src/code.c`

- Read the description and hint for each function.
- Implement the function body.
- Do **not** modify function signatures.

Part 2 — `src/tests.c`

- Read the description for each test function.
 - Implement the test body using `TEST_ASSERT_TRUE_MESSAGE`.
 - Use only **stack variables** — no `malloc`.
 - Do **not** modify function names or signatures.
6. When all local tests pass — **commit** and **push** to your repository.
 7. Go to the **Actions** tab on GitHub and verify the CI results.

Functions to implement — `src/code.c`

Function	Description
<code>swap</code>	Swap values of two integers via pointers
<code>find_last_char</code>	Return pointer to last character before <code>\0</code>
<code>link_three</code>	Connect three nodes: <code>a -> b -> c -> NULL</code>
<code>remove_middle</code>	Reconnect <code>a -> c</code> , disconnect <code>b</code>
<code>remove_last</code>	Set <code>b->nextPtr = NULL</code>

Function	Description
<code>remove_first</code>	Set <code>a->nextPtr = NULL</code>
<code>swap_ptrs</code>	Swap two pointers using double pointers
<code>nullify</code>	Set pointer to NULL via double pointer
<code>assign_bytes</code>	Assign bytes 1..8 via cast pointer
<code>sum_chain</code>	Return sum of all values in chain

Tests to implement — `src/tests.c`

Test function	What to verify
<code>test_swap_basic</code>	Two different values are exchanged
<code>test_swap_equal</code>	Equal values remain unchanged
<code>test_find_last_basic</code>	Correct pointer to last character
<code>test_find_last_single</code>	Single character string
<code>test_find_last_empty</code>	Empty string returns NULL
<code>test_link_three_chain</code>	All three pointers correctly set
<code>test_remove_middle_reconnects</code>	<code>a->nextPtr == &c, b->nextPtr == NULL</code>
<code>test_remove_last_truncates</code>	<code>b->nextPtr == NULL</code>
<code>test_remove_first_isolates</code>	<code>a->nextPtr == NULL</code> , rest intact
<code>test_swap_ptrs_redirects</code>	Each pointer points to the other variable
<code>test_swap_ptrs_values_unchanged</code>	Original integer values not modified
<code>test_nullify_sets_null</code>	Pointer is NULL after call
<code>test_nullify_value_unchanged</code>	Original value not modified
<code>test_assign_bytes_first_and_last</code>	Byte 0 == 1, byte 7 == 8
<code>test_assign_bytes_all</code>	Every byte <code>i</code> contains <code>i+1</code>
<code>test_sum_chain_basic</code>	Correct sum for three nodes
<code>test_sum_chain_single</code>	Single node returns its value
<code>test_sum_chain_null</code>	NULL returns 0

Rules

- Work on **stack only** — no `malloc`, no `free`
 - Do **not** modify function signatures in `code.c` or `tests.c`
 - Every `TEST_ASSERT_TRUE_MESSAGE` must include a meaningful message
 - CI runs your `tests.c` together with instructor tests — both must pass
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Grading

Component	Points
All functions correctly implemented	5
All test functions correctly written	5
Total	10

Author block — required

```
char *AUTHOR_NAME      = "Your Name";  
char *AUTHOR_AUTHORSHIP = "..."
```